

Daily Progress Report

Week 5_Day5: Dynamic Routing & URL Parameters

Program	3-Month MERN Stack Internship Program
Intern Name	Mohsin Ali
Week	Week_5 (Day 5)
Date	25 July, 2026

Day 5: Dynamic Routing & URL Parameters

1.1 Topics Studied

- Understanding dynamic routing in React using URL parameters
- Creating parameterized routes with react-router-dom (e.g., /products/:id)
- Accessing route parameters using the useParams hook
- Displaying collections of data using reusable React components
- Rendering dynamic lists using the map() method with unique keys
- Navigating between pages using Link components
- Building detail pages that display content based on URL parameters
- Handling invalid product IDs gracefully with conditional rendering
- Extending a modular React application by integrating additional pages into the existing routing architecture
- Designing responsive grid layouts and detail pages using modern UI principles

1.2 Task Description

Extended the Day 4 multi-page React application by introducing a **Products** section that demonstrates dynamic routing using URL parameters. A dedicated Products page displays a responsive grid of product items, each containing essential information and a navigation link. Selecting a product routes the user to a unique detail page using a parameterized URL (e.g., /products/:id). The application retrieves the corresponding product information based on the URL parameter and displays its complete details. Invalid product IDs are handled gracefully by displaying an informative message, ensuring a smooth user experience while maintaining the modular project structure established in the previous task.

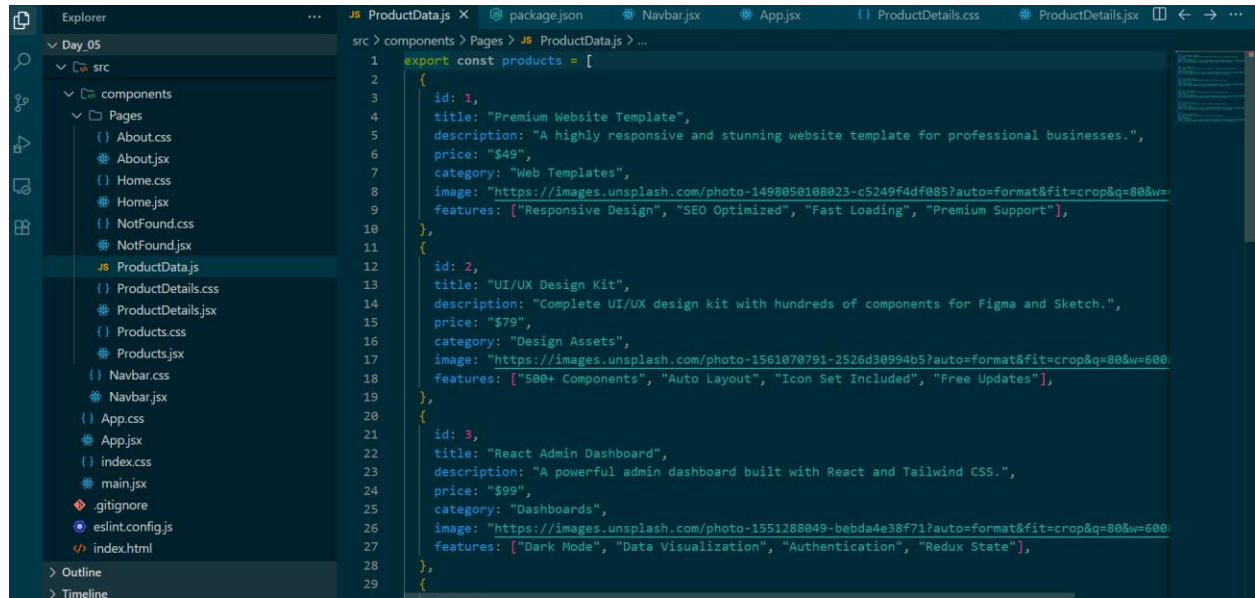
1.3 Implementation Details

- Extended the existing routing configuration by adding a Products page.
- Configured a dynamic route using Route with the path /products/:id.
- Implemented the useParams hook to retrieve the product ID from the URL.
- Created a responsive Products page that displays a grid of product cards.
- Rendered the product list dynamically using the map() method with unique keys.
- Added Link components to each product card for seamless navigation to its respective detail page.
- Developed a reusable Product Details page that displays information based on the selected product ID.
- Implemented conditional rendering to display an appropriate message when an invalid or non-existent product ID is accessed.
- Preserved client-side navigation without full-page reloads, ensuring a fast and responsive user experience.
- Maintained the modular folder structure by organizing pages, components, routing logic, and styling files separately for improved maintainability and scalability.
- Applied responsive layouts and modern CSS styling to ensure consistent user experience across desktop and mobile devices.

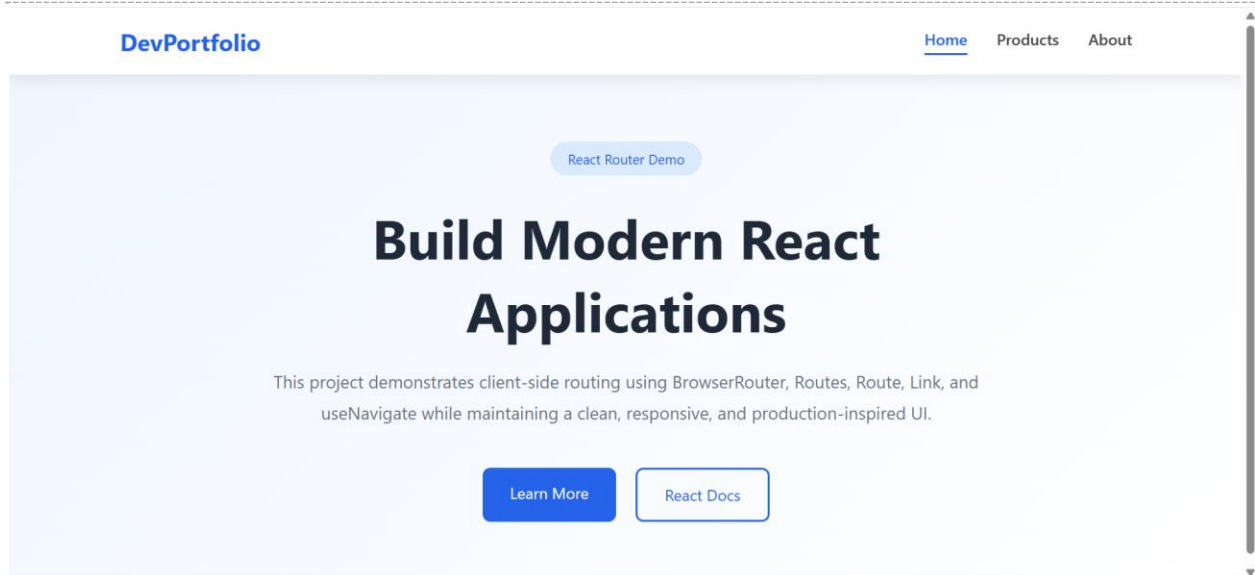
1.4 Outcome

Successfully extended the existing React routing application by implementing dynamic routing with URL parameters. The project strengthened understanding of configuring parameterized routes, accessing route parameters using useParams, navigating to dynamic pages through Link, and displaying data based on URL values. Additionally, handling invalid routes gracefully, organizing the application into reusable components, and maintaining a scalable project architecture reinforced practical knowledge of modern React development and real-world client-side routing techniques.

Screenshots:



Screenshots:





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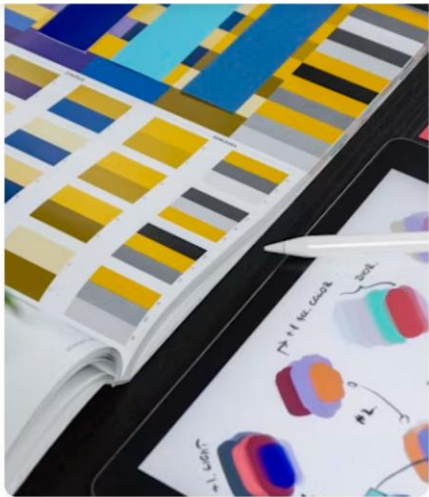


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